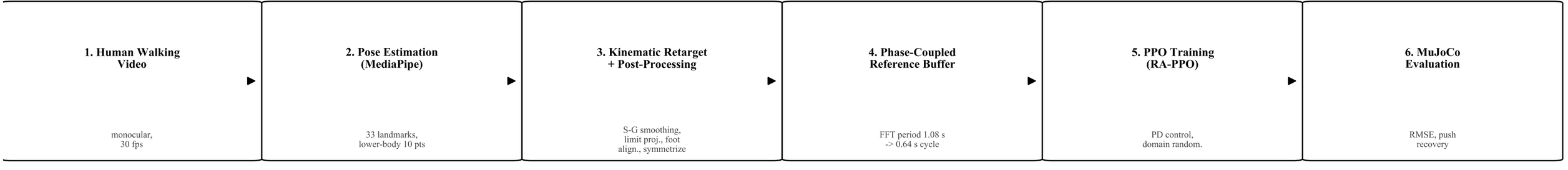
1. Human Walking Video
monocular, 30 fps
2. Pose Estimation (MediaPipe)
33 landmarks, lower-body 10 pts
3. Kinematic Retarget + Post-Processing
S-G smoothing, limit proj., foot align., symmetrize
4. Phase-Coupled Reference Buffer
FFT period 1.08 s -> 0.64 s cycle
5. PPO Training (RA-PPO)
PD control, domain random.
6. MuJoCo Evaluation
RMSE, push recovery